

# Summary-based post GWAS analysis

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# Lecture Rmd files

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<https://github.com/stat540-UBC/lectures>

# Today's lecture

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- 1 Summary-based analysis (LD score regression)
- 2 Stratified LD score regression
- 3 A brief introduction to causal inference
- 4 Mendelian Randomization and TWAS

# Notations

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- $X: n \times p$  genotype matrix of  $n$  samples/individuals and  $p$  variants
- $Y: n \times 1$  phenotype vector of  $n$  samples/individuals
- $z: p \times 1$  summary z-scores



# Using the 1000 Genomes data via bigsnpr

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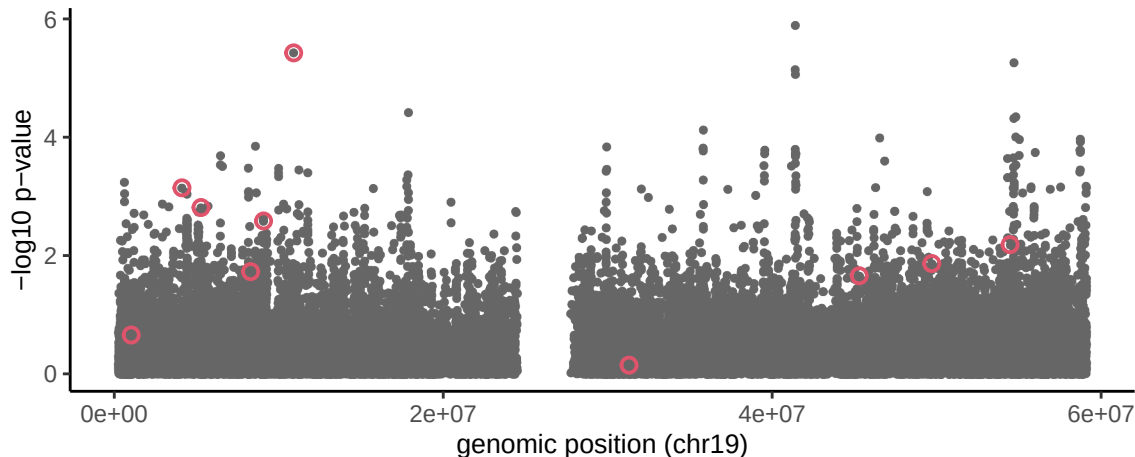
```
library(bigsnp)  
download_1000G("../data/genotype/")
```

```
data <- snp_attach(.bk.file)  
str(data, max.level=1, strict.width = "cut")
```

```
## List of 3  
## $ genotypes:Reference class 'FBM.code256' [package "bigstatsr"] with 16 fields  
## ..and 26 methods, of which 12 are possibly relevant  
## $ fam      : 'data.frame': 2490 obs. of 6 variables:  
## $ map      : 'data.frame': 1664852 obs. of 6 variables:  
## - attr(*, "class")= chr "bigSNP"
```

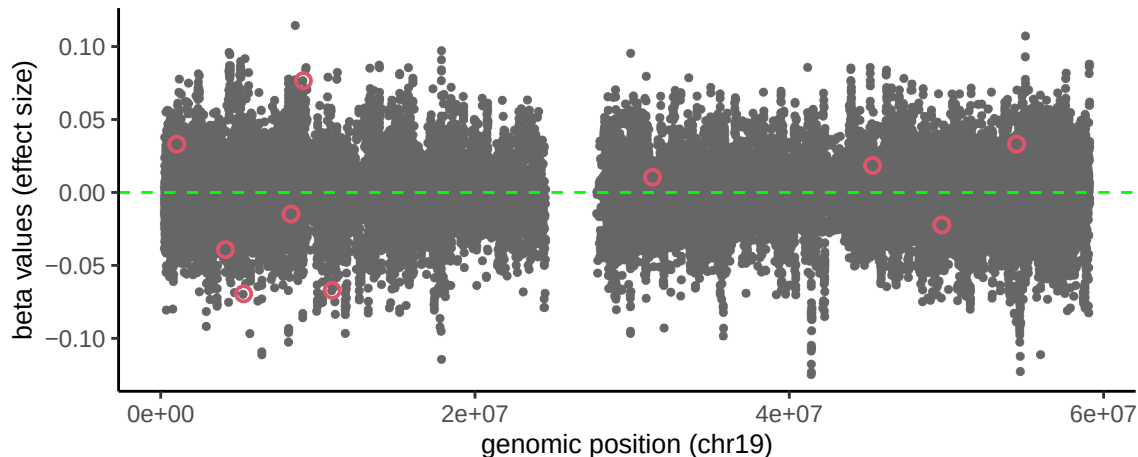
# We may not have a full data (especially phenotypes)

Instead, we could have just summary statistics (e.g., Manhattan plot).



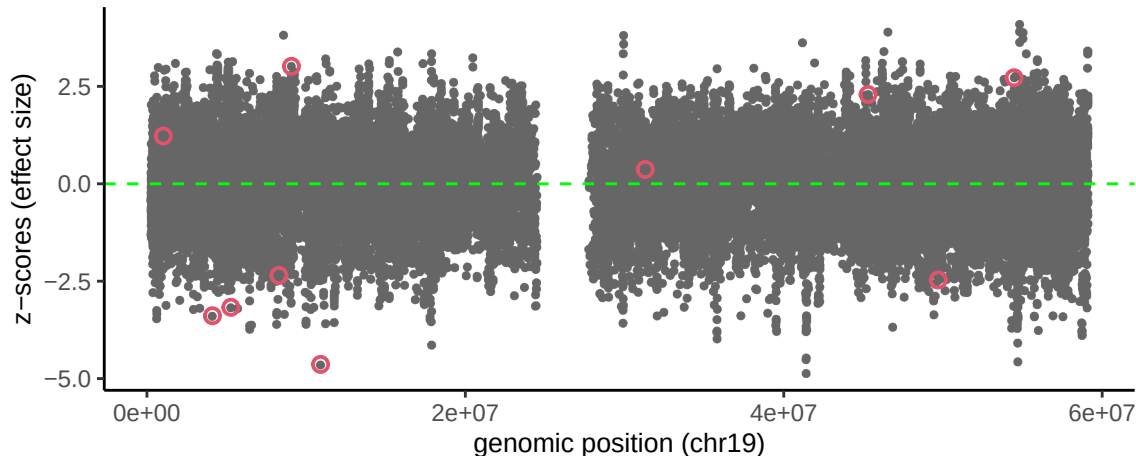
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# We could have full access to univariate stats

---

```
head(sumstat.dt[,.(chromosome, marker.ID, physical.pos, beta, se, pv)])
```

| ##    | chromosome | marker.ID | physical.pos | beta         | se         | pv        |
|-------|------------|-----------|--------------|--------------|------------|-----------|
| ##    | <int>      | <char>    | <int>        | <num>        | <num>      | <num>     |
| ## 1: | 19         | rs8100066 | 260912       | -0.013392212 | 0.03209562 | 0.6765244 |
| ## 2: | 19         | rs8102615 | 260970       | -0.038096443 | 0.02945621 | 0.1960189 |
| ## 3: | 19         | rs8105536 | 261033       | -0.014187124 | 0.03210357 | 0.6585875 |
| ## 4: | 19         | rs2312724 | 266034       | -0.004839159 | 0.01335465 | 0.7171156 |
| ## 5: | 19         | rs4897933 | 266888       | 0.003032540  | 0.01061138 | 0.7750691 |
| ## 6: | 19         | rs1020382 | 267039       | -0.007183893 | 0.03202064 | 0.8225019 |

# Figure out how these univariate stats were calculated

---

A variant-by-variant model:

$$Y_i \sim X_{ij}\beta_j + \epsilon, \quad \epsilon \sim \mathcal{N}(0, \sigma^2)$$

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A variant-by-variant model:

$$Y_i \sim X_{ij}\beta_j + \epsilon, \quad \epsilon \sim \mathcal{N}(0, \sigma^2)$$

Summary statistics for each variant  $j$ :

$$\hat{\beta}_j = \frac{\sum_i X_{ij}Y_i}{\sum_i X_{ij}^2}, \quad \hat{V}[\beta_j] = \frac{\hat{\sigma}_\epsilon^2}{\sum_{i=1} X_{ij}^2}, \quad Z_j = \frac{\hat{\beta}_j}{\sqrt{\hat{V}[\beta_j]}}$$

# How did we derive these results?

---

$$\mathcal{L} = \frac{1}{2} \sum_{i=1}^n (Y_i - X_{ij}\beta)^2$$



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$$\mathcal{L} = \frac{1}{2} \sum_{i=1}^n (Y_i - X_{ij}\beta)^2$$

$$\frac{\partial \mathcal{L}}{\partial \beta} = \sum_i (Y_i - X_{ij}\beta) X_{ij} = 0$$

$$\sum_i X_{ij}^2 \beta = \sum_i Y_i X_{ij}$$

$$\beta = \sum_i Y_i X_{ij} / \sum_i X_{ij}^2$$

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$$\mathcal{L} = \frac{1}{2} \sum_{i=1}^n (Y_i - X_{ij}\beta)^2$$

$$\begin{aligned} \frac{\partial^2 \mathcal{L}}{\partial \beta^2} &= - \sum_i X_{ij}^2 \\ &= -1/\hat{\sigma}_\beta^2 \\ \hat{\sigma}_\beta^2 &= 1/\sum_i X_{ij}^2 \end{aligned}$$

# A very rough idea of how we can derive

---

$$\begin{aligned}\mathcal{L} &= \ln \mathcal{N}(\beta|\mu, \sigma^2) \\ &= -\frac{1}{2} \ln \sigma^2 - \frac{1}{2\sigma^2}(\beta - \mu)^2 + \dots \\ \frac{\partial \mathcal{L}}{\partial \beta} &= -\frac{1}{\sigma^2}(\beta - \mu) = 0 \\ \frac{\partial^2 \mathcal{L}}{\partial \beta^2} &= -\frac{1}{\sigma^2}\end{aligned}$$

# We can easily obtain GWAS summary statistics data

---

$$\hat{\beta}_j = \frac{\sum_i X_{ij} Y_i}{\sum_i X_{ij}^2}, \quad \hat{V}[\beta_j] = \frac{\hat{\sigma}_\epsilon^2}{\sum_{i=1} X_{ij}^2}$$

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$$\hat{\beta}_j = \frac{\sum_i X_{ij} Y_i}{\sum_i X_{ij}^2}, \quad \hat{V}[\beta_j] = \frac{\hat{\sigma}_\epsilon^2}{\sum_{i=1} X_{ij}^2}$$

$$Z_j = \frac{\hat{\beta}_j}{\sqrt{\hat{V}[\beta_j]}}, \quad \chi_j^2 = Z_j^2$$

# Underlying assumption: Our trait of interest is polygenic

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Thus, we assume there are multivariate effects

$$Y_i = \sum_{j=1}^p X_{ij} \theta_j + \epsilon, \quad \forall i \in [n], \text{ or}$$

$$y = X\theta + \epsilon,$$

# Revised problem definition of GWAS

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## Input

- GWAS Z scores  $p \times 1$
- $p \times p$  LD (covariance) matrix
- (or) ref.  $n \times p$  genotype  $X$

## Output

- A polygenic regression model

$$y \sim \sum_j X_{ij} \theta_j$$

Is univariate  $\beta_j$  the same as multivariate  $\theta_j$ ?



# Goal: How can we recover the multivariate one?

---

Data:

$$\hat{Z}_j = \sum_{i=1}^n X_{ij} Y_i / \sigma_{\epsilon} \sqrt{n}$$

for all  $j \in [p]$ .

Goal: Recover  $p \times 1$  multivariate effect size vector  $\theta$ .

# Inference $\approx$ reversing the generative process

---

Data:

$$\hat{Z}_j = \sum_{i=1}^n X_{ij} Y_i / \sigma_\epsilon \sqrt{n}$$

for all  $j \in [p]$ .

Hormozdiari *et al.*, Genetics (2014); Zhu and Stephens, Annals of Applied Statistics (2017)

# Inference $\approx$ reversing the generative process

---

With a matrix-vector notation (to save space),

$$\begin{matrix} p \times 1 & \mathbf{z} & \\ \text{univariate} & & \end{matrix} = \frac{1}{\sigma\sqrt{n}} X^\top \begin{matrix} \mathbf{y} \\ n \times 1 \text{ phenotype} \end{matrix}$$

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# Inference $\approx$ reversing the generative process

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$$\begin{aligned} \underbrace{p \times 1}_{\text{univariate}} \mathbf{z} &= \frac{1}{\sigma\sqrt{n}} X^\top \underbrace{\mathbf{y}}_{n \times 1 \text{ phenotype}} = \frac{1}{\sigma\sqrt{n}} X^\top \underbrace{(X\theta + \epsilon)}_{\text{a multivariate model}} \\ &= \frac{\sqrt{n}}{\sigma} \underbrace{\left( \frac{1}{n} X^\top X \right)}_{\text{LD}} \theta + \frac{1}{\sigma\sqrt{n}} X^\top \epsilon \end{aligned}$$

Hormozdiari *et al.*, Genetics (2014); Zhu and Stephens, Annals of Applied Statistics (2017)

# Inference $\approx$ reversing the generative process

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$$\begin{aligned} \underset{p \times 1 \text{ univariate}}{\mathbf{z}} &= \frac{1}{\sigma\sqrt{n}} \mathbf{X}^\top \underset{n \times 1 \text{ phenotype}}{\mathbf{y}} = \frac{1}{\sigma\sqrt{n}} \mathbf{X}^\top \underbrace{(\mathbf{X}\boldsymbol{\theta} + \boldsymbol{\epsilon})}_{\text{a multivariate model}} \\ &= \mathbf{R} \frac{\sqrt{n}}{\sigma} \boldsymbol{\theta} + \tilde{\boldsymbol{\epsilon}}, \quad \tilde{\boldsymbol{\epsilon}} \sim \mathcal{N}(\mathbf{0}, \mathbf{R}) \end{aligned}$$

where  $\mathbf{R} = n^{-1} \mathbf{X}^\top \mathbf{X}$  is an empirical LD matrix.

Hormozdiari *et al.*, Genetics (2014); Zhu and Stephens, Annals of Applied Statistics (2017)

# Running SuSiE in a summary statistics mode

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The underlying model:

$$\mathbf{z} \sim \mathcal{N}\left(\frac{\sqrt{n}}{\sigma} R \boldsymbol{\theta}, R\right)$$

```
library(susieR)
z <- sumstat.dt$beta/sumstat.dt$se
R <- cov(X)
susie.out <- susie_rss(z, R, n=nrow(X))
```

# Running SuSiE in a summary statistics mode

The underlying model:

$$\underset{\text{univariate } z}{\mathbf{z}} \sim \mathcal{N} \left( \frac{\sqrt{n}}{\underbrace{\sigma}_{\text{LD}}} R \underbrace{\theta}_{\text{multivariate effect}}, \underbrace{R}_{\text{LD}} \right)$$

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library(susieR)
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# LD-score: another type of summary-based analysis

---

What is a generative model for a  $\chi_j^2$ , which is ( $= Z_j^2$ ), statistics vector?

We have seen this relationship in the fine-mapping model:

$$\underset{\text{univariate, summary stat}}{Z_j} = \frac{\sqrt{n}}{\sigma} \sum_k \underset{\text{LD between j and k}}{R_{jk}} \underset{\text{multivariate, true effect}}{\theta_k} + \epsilon_j$$

where  $\epsilon \sim \mathcal{N}(0, 1)$ .

# LD-score: another type of summary-based analysis

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What is a generative model for a  $\chi_j^2$ , which is ( $= Z_j^2$ ), statistics vector?

$$\mathbb{E}[\chi_j^2] = \mathbb{E}[Z_j^2] = \mathbb{E}\left(\sqrt{n} \sum_k R_{jk} \theta_k + \epsilon_j\right)^2$$

Bulik-Sullivan *et al.*, *Nature Genetics* (2014); Finucane *et al.*, *Nature Genetics* (2015)

# LD-score: another type of summary-based analysis

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$$\mathbb{E}[\chi_j^2] = \mathbb{E}[Z_j^2] = \mathbb{E} \left( \sqrt{n} \sum_k R_{jk} \theta_k + \epsilon_j \right)^2$$

If the effects are independent of each other, i.e.,  $\mathbb{E}[\theta_k \theta_j] = 0$  for all  $k \neq j$ ,

$$\mathbb{E}[\chi_j^2] = n \underbrace{\sum_k R_{jk}^2}_{\text{LD-score}} \mathbb{E}[\theta_k^2] + 1$$

# You may calculate LD-scores on genotype bed file

---

```
library(bigsnpR)
ld.scores <-
  bed_ld_scores(
    bed(.bed.file),
    size = 500,
    ncores = 8)
```

# Baseline LD-score to measure polygenic heritability

---

Assuming that all the variants equally contribute,

$$\mathbb{E}[\theta_k^2] = \tau/p,$$

where  $p$  is the total number of SNPs,

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We get

$$\mathbb{E}[\chi_j^2] = n \underbrace{\sum_k R_{jk}^2}_{\text{LD-score}} \mathbb{E}[\theta_k^2] + 1 = \underbrace{n}_{\text{sample size}} \underbrace{l_j}_{\text{LD score}} \underbrace{\frac{\tau}{p}}_{\text{per SNP heritability}} + 1$$

where  $p$  is the total number of SNPs.



# Baseline LD-score to measure polygenic heritability

We can treat the relationships as a regression model and find the heritability parameters by regressing the observed  $\chi^2$  statistics on the reference LD scores  $l_j$ :

$$\begin{pmatrix} \chi_1^2 \\ \vdots \\ \chi_j^2 \\ \vdots \end{pmatrix} \sim \frac{n}{p} \begin{pmatrix} l_1 \\ \vdots \\ l_j \\ \vdots \end{pmatrix} \text{ per SNP } \tau \text{ heritability} + n\phi \text{ genomic inflation} + 1 \text{ null}$$

If the intercept of  $\{\chi_j^2\}$  deviate from 1, we can interpret that the GWAS statistics are inflated by some unadjusted population structures or other confounding factors.

Bulik-Sullivan *et al.*, *Nature Genetics* (2014)

# What we need to know in this section

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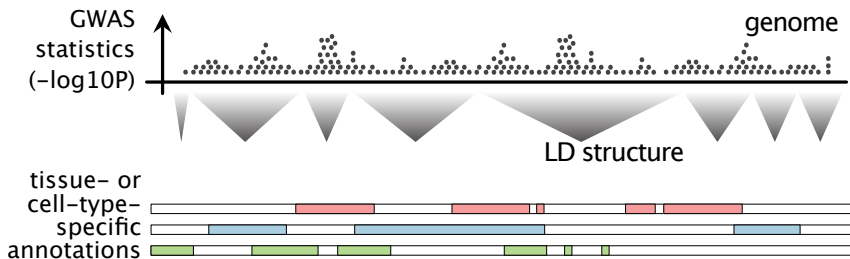
- ① GWAS summary statistics contain good amount of information
- ② In the summary data, we need to deal with LD structure  $R$

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# Stratified LD-score regression to partition (stratify) total heritability into multiple genomic annotations



E.g., How much heritability of a disease is explained by tissue-specific epigenomic signals?

# Stratified LD-score regression in math

---

When genome is partitioned by annotations (e.g., epigenetic tracks)

$$\mathbb{E}[\chi_j^2] = \frac{n}{p} \sum_t l_{jt} \tau_t + n\phi + 1$$

stratified heritabilitygenomic inflationnull

where we use partitioned LD-scores for each annotation type  $t$

$$l_{jt} = \sum_k R_{jk}^2 I\{k \in \mathcal{A}_t\}.$$

# Stratified LD-score regression in math

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$$\mathbb{E}[\chi_j^2] = \frac{n}{p} \sum_t l_{jt} \tau_t + n\phi + 1$$

stratified heritabilitygenomic inflationnull

where we use partitioned LD-scores for each annotation type  $t$

$$l_{jt} = \sum_k R_{jk}^2 I\{k \in \mathcal{A}_t\}.$$

Instead of assuming a single parameter for the overall per-SNP heritability  $\tau$ , we can “partition” this total heritability into annotation-type-specific ones,  $\{\tau_t\}$ .

# Stratified LD-score regression in math

When genome is partitioned by annotations (e.g., epigenetic tracks)

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stratified heritabilitygenomic inflationnull

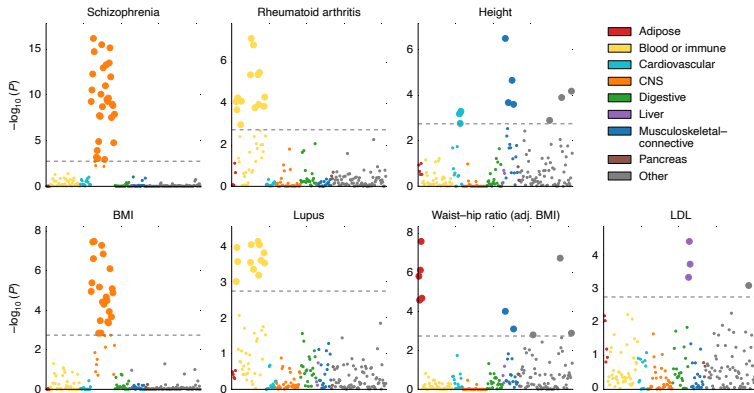
where we use partitioned LD-scores for each annotation type  $t$

$$l_{jt} = \sum_k R_{jk}^2 I\{k \in \mathcal{A}_t\}.$$

$$\begin{pmatrix} \chi_1^2 \\ \vdots \end{pmatrix} \sim \frac{n}{p} \begin{pmatrix} l_{11} & l_{12} & l_{1t} & \cdots \\ \vdots & \vdots & \vdots & \vdots \end{pmatrix} \begin{pmatrix} \tau_1 \\ \vdots \end{pmatrix} + n\phi + 1$$

stratified LD scoresstratified heritabilitygenomic inflationnull

# Stratified LD-score regression can identify tissue-specific heritability enrichment



Finucane et al., *Nature Genetics* (2018)



# Bivariate LD-score regression

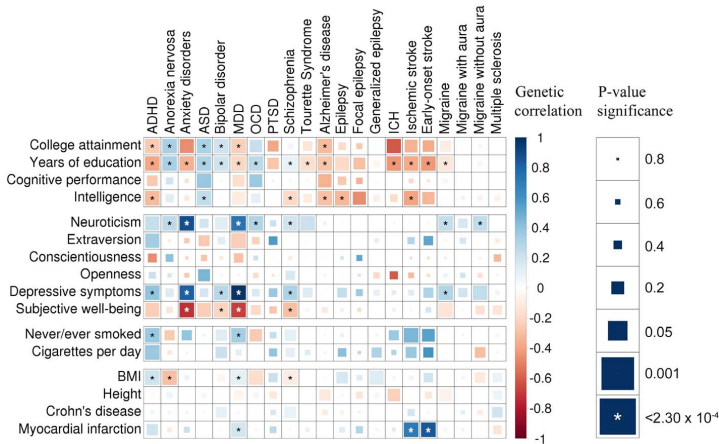
Instead of one  $\chi^2$  vector, we need to deal with the element-wise product of two vectors of z-scores (between a trait 1 and 2):

$$\begin{pmatrix} z_1^{(1)} z_1^{(2)} l_1 \\ \vdots \\ z_j^{(1)} z_j^{(2)} l_j \\ \vdots \end{pmatrix} \sim \frac{\sqrt{N_1 N_2}}{p} \begin{pmatrix} l_1 \\ \vdots \\ l_j \\ \vdots \end{pmatrix} \underbrace{\rho}_{\text{genetic correlation}} + \underbrace{\frac{\rho_0 N_s}{\sqrt{N_1 N_2}}}_{\text{sample sharing}}$$

where  $N_1$  and  $N_2$  count sample size of the GWAS 1 and 2;  $N_s$  is the number of control individuals shared between the two traits.

Bulik-Sullivan *et al.*, *Nature Genetics* (2015)

# Bivariate LD-score regression to test genetic correlations



The Brainstorm Consortium, *Science* (2018)

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# The goal of causal effect inference

---

- 1 Instead of "seeing" what happened (data analysis), we want to measure the effect of "doing" an experiment (intervention).

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- 2 Because correlation is not causation.

# The goal of causal effect inference

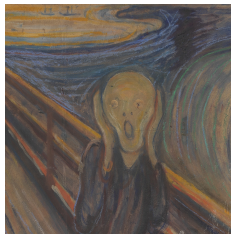
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- 1 Instead of "seeing" what happened (data analysis), we want to measure the effect of "doing" an experiment (intervention).
- 2 Because correlation is not causation.
- 3 If it is not causation, then what is it?

# Common misconception - correlation implies causation

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Henry Niles also wrote, “To contrast ‘causation’ and ‘correlation’ is unwarranted because causation is simply perfect correlation”



Pearl, *The Book of Why*, p.78

- How do we measure causal effects? Will any experiment do?

# A working example: Why correlation is not causation

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- $U_i$ : The age at which a person  $i$  enrolled to this study.
- $X_i$ : An indicator variable for a physician decided to administer this new preventative drug  $X_i = 1$  or not  $X_i = 0$ .
- $Y_i$ : Biomarker protein concentration (e.g., Amyloid-beta protein disrupts normal brain functions).



# A working example: How the data had been generated.

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- A person  $i$ 's age  $U_i$ .
- ① Prescribe the preventative drug based on the subject's age  $U_i$  (as they might need one):

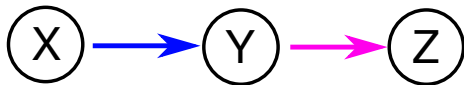
$$P(X_i = 1|U_i) = \sigma(U_i\delta)$$

- ② A biomarker protein concentration:

$$\mathbb{E}[Y_i|X, U] = X_i\tau + U_i\beta$$

# Directed Acyclic Graph as a language of causal inference

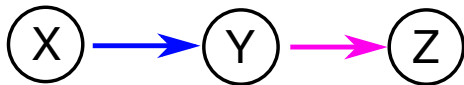
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$$p(X, Y, Z) = \underbrace{p(Y|X)}_{\text{first edge}} \underbrace{p(Z|Y)}_{\text{second edge}} \underbrace{p(X)}_{\text{first node}}$$

# DAG: Causal path/trail and reachability

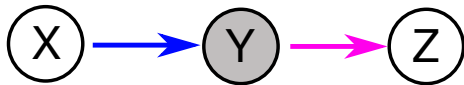
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- There is a path between  $X$  and  $Y$
- Also between  $Y$  and  $Z$
- Also from  $X$  to  $Z$

# DAG: Conditioning (and/or adjustment)

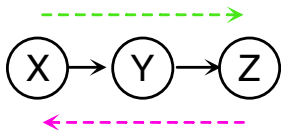
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- A shaded node = conditioning like  $P(Z|Y = y^*)$  and  $P(Y = y^*|X)$
- There is a path between  $X$  and  $Y$
- Also between  $Y$  and  $Z$
- But no path from  $X$  to  $Z$

# d-separation: testing conditional independence (flow vs. no flow)

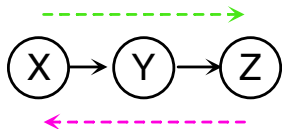
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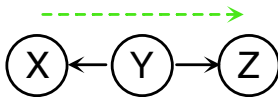
**Flow:** The outcome of  $Z$  depends on the effect of  $X$

# d-separation: testing conditional independence (flow vs. no flow)

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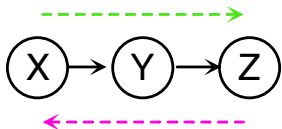


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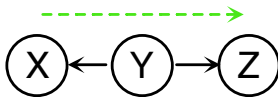


**Flow:** The outcome of  $Z$  depends on the effect of  $Y$ ; so does the outcome of  $X$  on that of  $Y$

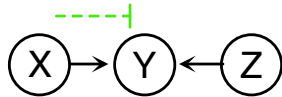
# d-separation: testing conditional independence (flow vs. no flow)



**Flow:** The outcome of  $Z$  depends on the effect of  $X$

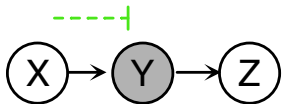
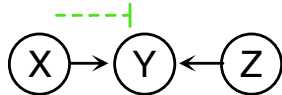
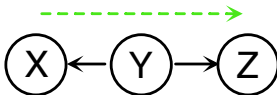
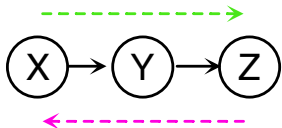


**Flow:** The outcome of  $Z$  depends on the effect of  $Y$ ; so does the outcome of  $X$  on that of  $Y$



**No Flow:** The outcome of  $Z$  only affects  $Y$ ; so does  $X$

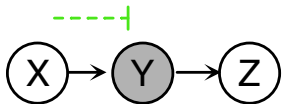
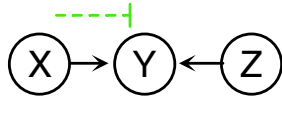
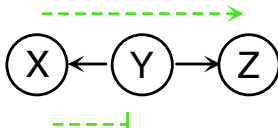
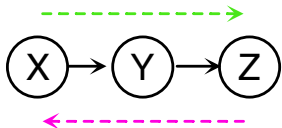
# d-separation: testing conditional independence (flow vs. no flow)



**No Flow:** The outcome of  $Z$  solely depends on  $Y = y^*$ , not  $X$



# d-separation: testing conditional independence (flow vs. no flow)

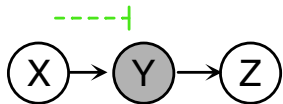
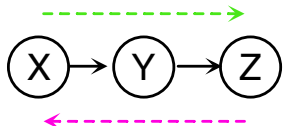


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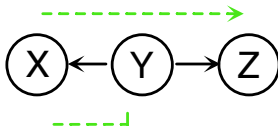


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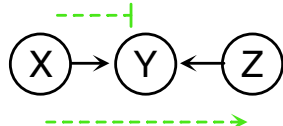
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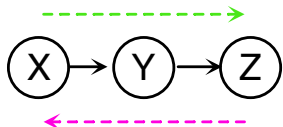


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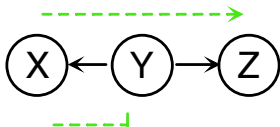


**Flow:** The observed outcome  $Y = y^*$  could stem from  $X$  or/and  $Z$

# d-separation: testing conditional independence (flow vs. no flow)



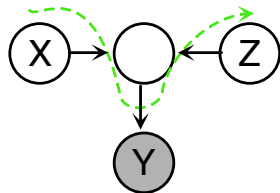
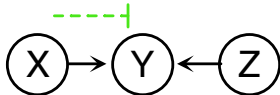
**No Flow:** The outcome of  $Z$  solely depends on  $Y = y^*$ , not  $X$



**No Flow:** The outcome of  $Z$  solely depends on  $Y = y^*$ , not  $X$



**Flow:** deeper v-structure



# Going back to the same working example

---

```
## Simulation parameters
n = 200; .delta = 2; .tau = -1; .beta = 2; .eps = .3
sgm <- function(x) 1/(1+ exp(-x))

## A generative scheme
uu <- rnorm(n) # standardized age
xx <- rbinom(n=n, size=1, prob=sgm(uu * .delta))
yy <- xx * .tau + uu * .beta + rnorm(n) * .eps
```

- The variable  $U$  confounds  $X$  and  $Y$ .
- How can we represent it graphically?

If there were no confounding:

```
yy.unc <- xx * .tau + rnorm(n) * .eps
```

# Going back to the same working example

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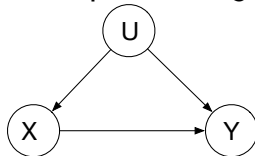
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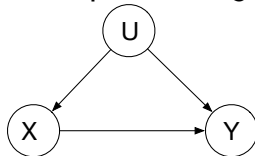
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- 1 How can we identify the causal effect,  $X \rightarrow Y$ ?
- 2 Why can't we just report the correlation/association between  $X$  and  $Y$ ?

# Going back to the same working example

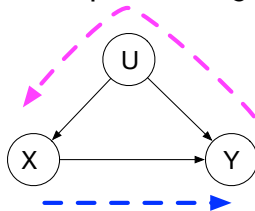
```
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n = 200; .delta = 2; .tau = -1; .beta = 2; .eps = .3
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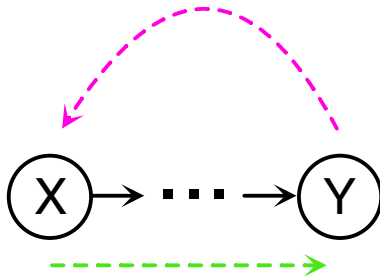
- 1 How can we identify the causal effect,  $X \rightarrow Y$ ?
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# It happened because

---

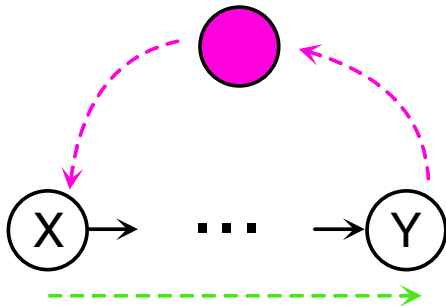
Question: Which nodes should be “conditioned” and/or “adjusted” to block a reverse path from  $Y$  to  $X$ ?



# It happened because

---

Question: Which nodes should be “conditioned” and/or “adjusted” to block a reverse path from  $Y$  to  $X$ ?

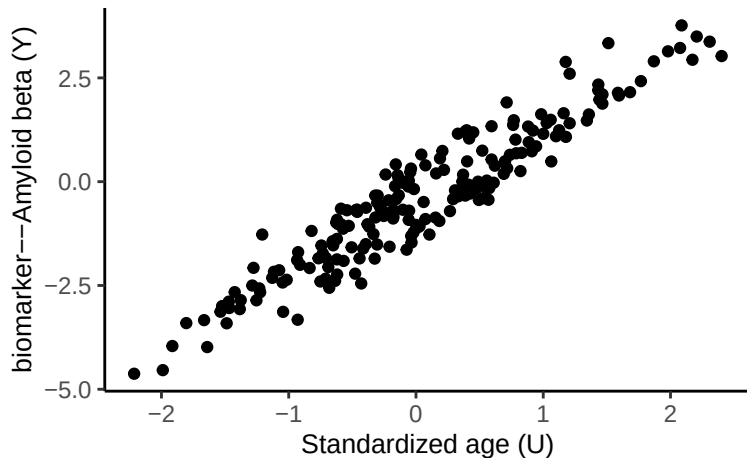


# Differential biomarker analysis suggests that our drug can exacerbate AD

---

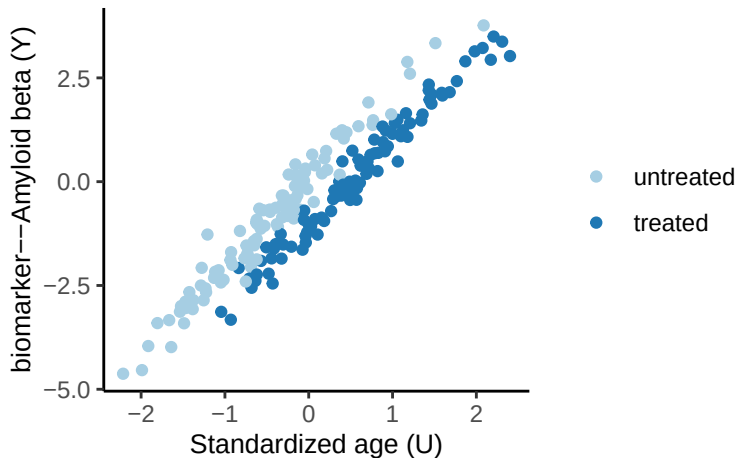


# Prevalence of AD increase with age



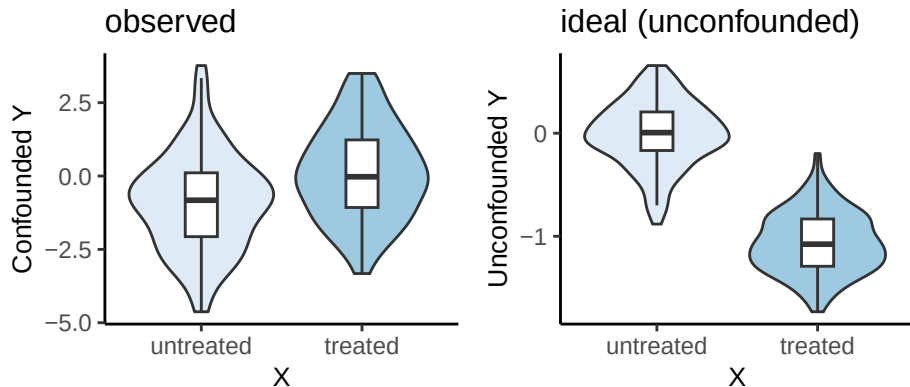
*Doctors could have prescribed our drug more frequently to older people because the prevalence of AD tends to increase with age.*

# Prevalence of AD increase with age



*Doctors could have prescribed our drug more frequently to older people because the prevalence of AD tends to increase with age.*

# Only if we could observe the unconfounded data



# What could have been an ideal experimental design?

---

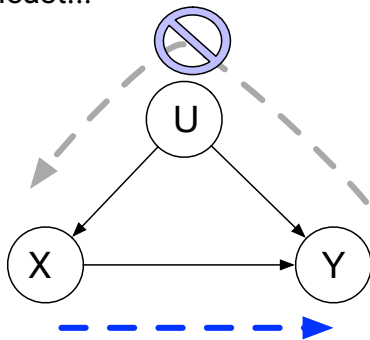
*There is no “backdoor” between a putative cause variable ( $X$ ) and outcome variable ( $Y$ ).*



# What could have been an ideal experimental design?

*There is no “backdoor” between a putative cause variable ( $X$ ) and outcome variable ( $Y$ ).*

At least...





# RCT: an ideal experiment (feat. RA Fisher)

---

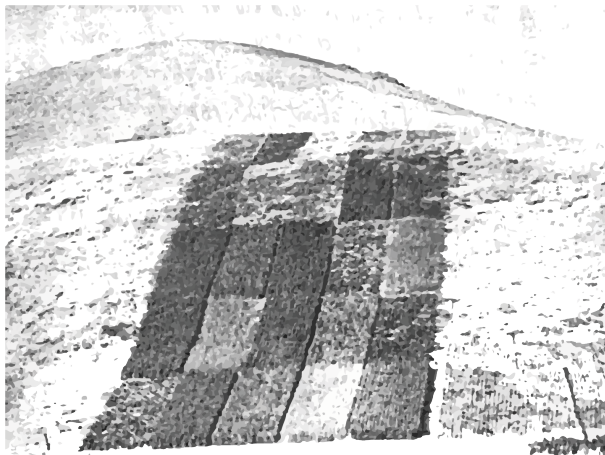
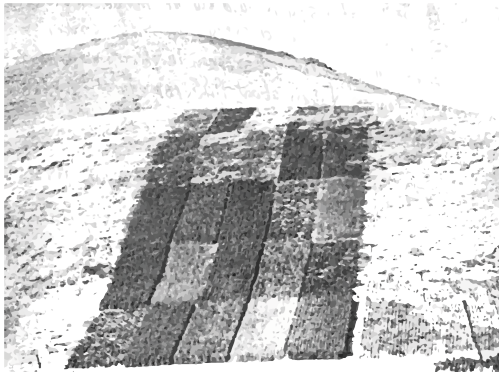


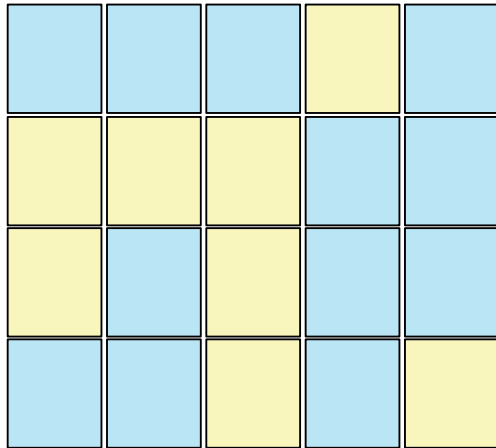
Image source: '[www.adelaide.edu.au](http://www.adelaide.edu.au)'

# RCT: an ideal experiment (feat. RA Fisher)

Crops grown in the plots

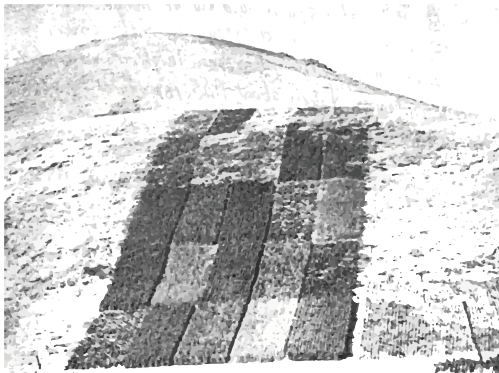


Experimental design

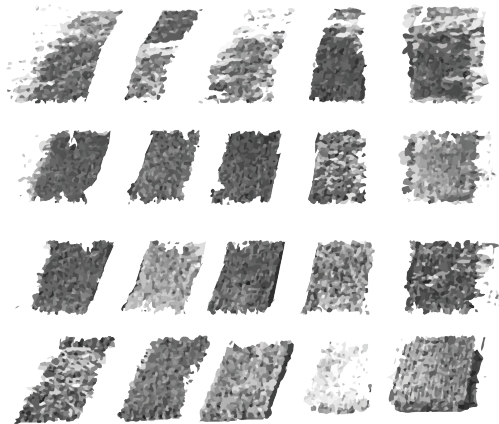


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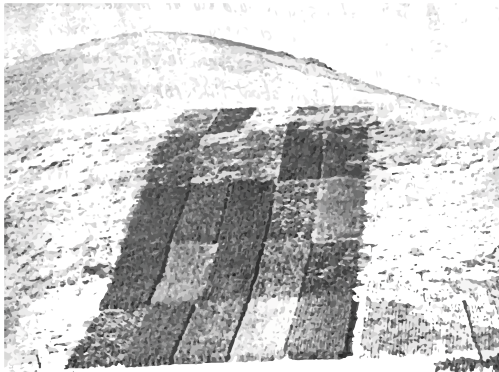


Random assignment of fertilizer

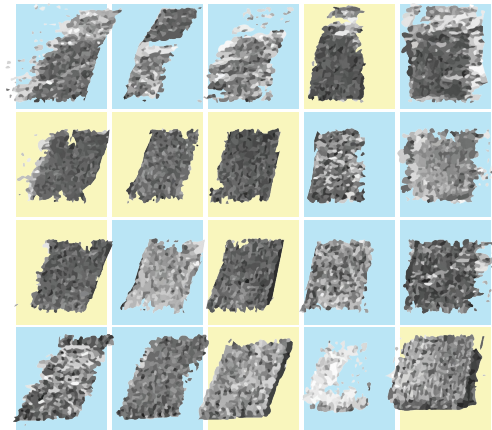


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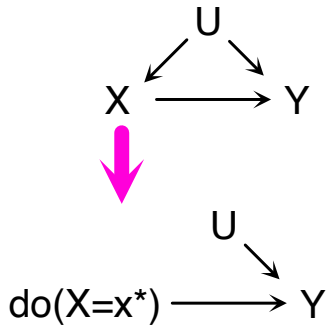


The treated vs. untreated

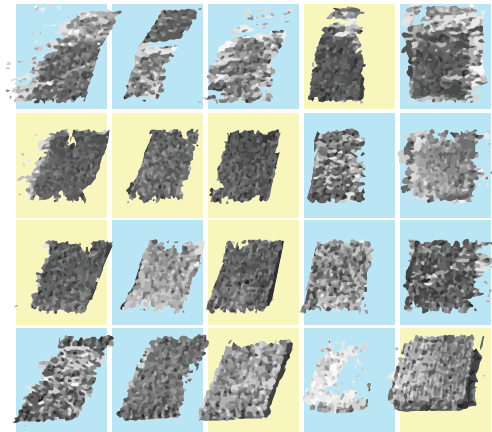


# RCT: an ideal experiment (feat. RA Fisher)

Randomized Control Trial  $\Rightarrow$  "doing"

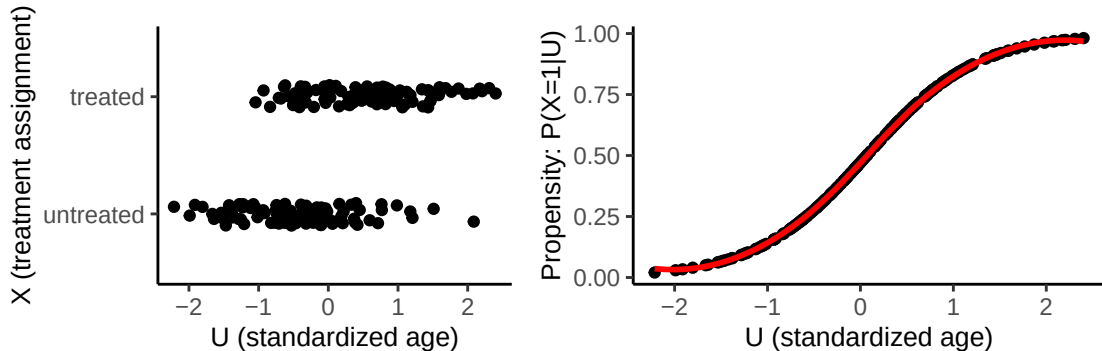


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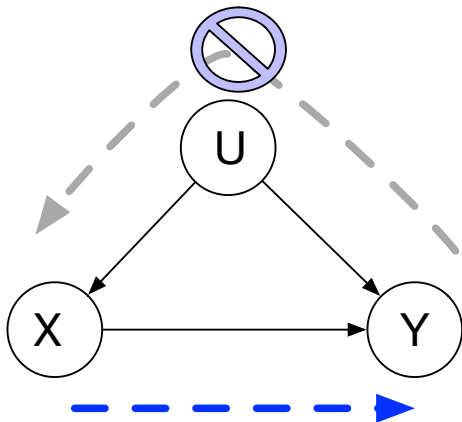


# Going back to the toy example: Can we save the study?

Had there been any systematic bias, thou shall adjust it.



# Inverse Propensity Weighting “reverse” confounding



```
p.xu <-  
  glm(x~u, family="binomial") %>%  
  predict() %>%  
  sigmoid() %>%  
  clamp() # avoid strict 0/1  
  
ww <- x / p.xu + (1-x) / (1-p.xu)
```

Propensity: Pr. of assignment  $X = 1$ :

$$\log \frac{p(X|U)}{1 - p(X|U)} \approx \beta_0 + \beta_1 U$$

# Inverse Propensity Weighting “reverse” confounding

$$\hat{Y}_i^{(1)} = \frac{X_i Y_i}{\hat{p}(X_i = 1 | U_i)}$$
$$\hat{Y}_i^{(0)} = \frac{(1 - X_i) Y_i}{1 - \hat{p}(X_i = 1 | U_i)}$$

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equivalently give weights for  $\forall i$

$$W_i \propto \begin{cases} 1/\hat{p}(X_i = 1 | U_i) & X_i = 1 \\ 1/\hat{p}(X_i = 0 | U_i) & X_i = 0 \end{cases}$$

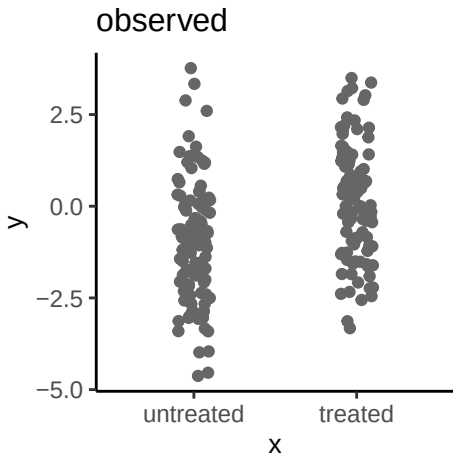
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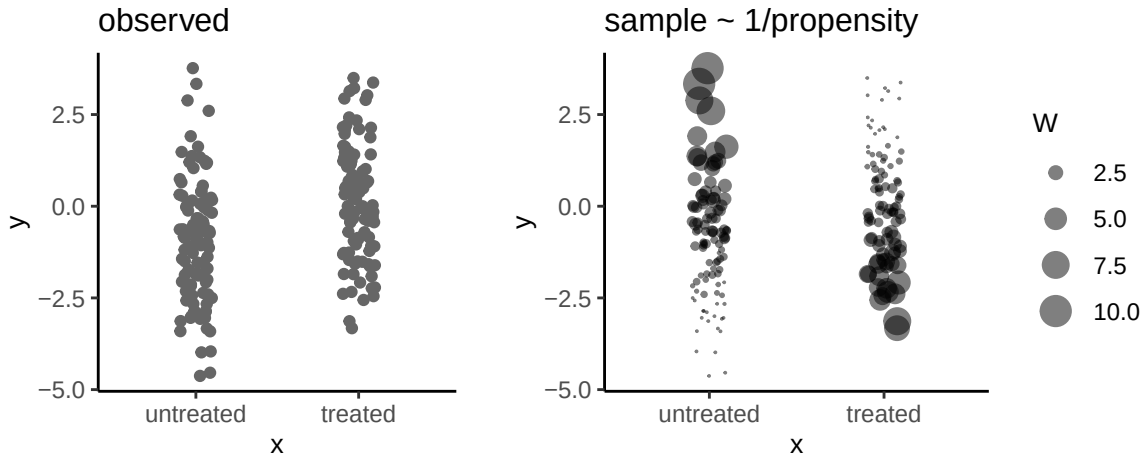
$$\log \frac{p(X|U)}{1 - p(X|U)} \approx \beta_0 + \beta_1 U$$

# Take samples inversely proportional to propensity

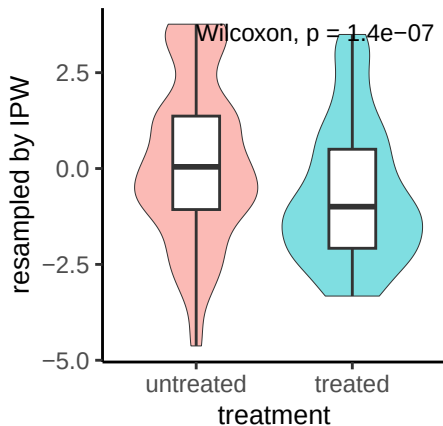
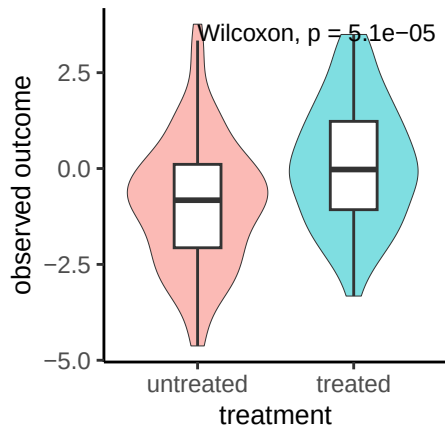
---



# Take samples inversely proportional to propensity



# If we bootstrap (sample with replacement)...



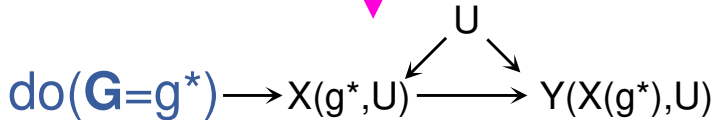
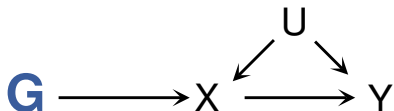
# Today's lecture

---

- 1 Summary-based analysis (LD score regression)
- 2 Stratified LD score regression
- 3 A brief introduction to causal inference
- 4 Mendelian Randomization and TWAS**

# Mendelian Randomization

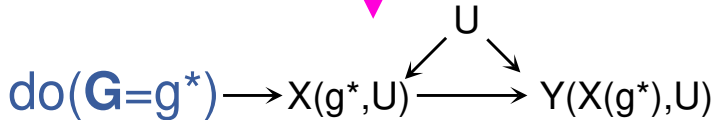
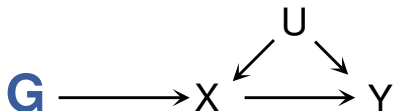
---



We will use a genetic variable  $G$  to mimic RCT

# Mendelian Randomization

---



If a genetic variable  $G$  is a valid instrumental variable...

# MR in modern epidemiology studies

---

- G: genotype
- X: APOE protein
- Y: cancer

$$G \rightarrow X \rightarrow Y$$

## APOLIPOPROTEIN E ISOFORMS, SERUM CHOLESTEROL, AND CANCER

SIR,—It is unclear whether the relation between low serum cholesterol levels and cancer<sup>1</sup> is causal. In many studies occult tumour may have depressed cholesterol levels though in others the relation was found when serum cholesterol had been measured many years before the cancer was diagnosed. The relation is probably not explained by diet, because in the Seven Countries Study cohorts with widely different diets and corresponding differences in mean cholesterol levels experienced similar mean cancer rates.<sup>2,3</sup> On the other hand, within each region cancer incidence was higher in men with a serum cholesterol in the lowest part of the cholesterol distribution for that country.<sup>3</sup> Thus, naturally low cholesterol levels are sometimes associated with increased cancer risk.<sup>1,3</sup>

Differences in the aminoacid sequence of apolipoprotein E (apo

Katan, *Lancet*, (1986)



# MR in modern epidemiology studies

- G: genotype
- X: APOE protein
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30TH THOMAS FRANCIS JR MEMORIAL LECTURE

**'Mendelian randomization': can genetic epidemiology contribute to understanding environmental determinants of disease?\***

George  
Davey Smith

George Davey Smith and Shah Ebrahim

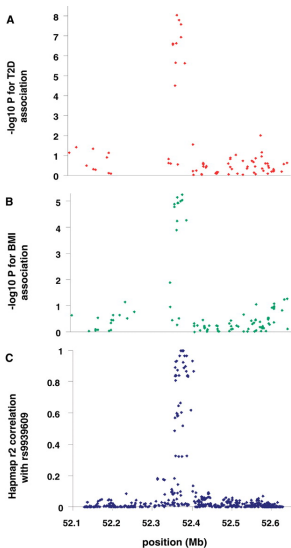
## **APOLIPOPROTEIN E ISOFORMS, SERUM CHOLESTEROL, AND CANCER**

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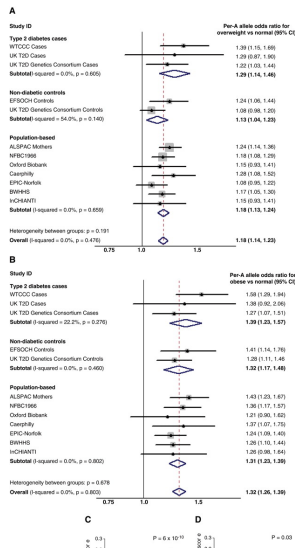
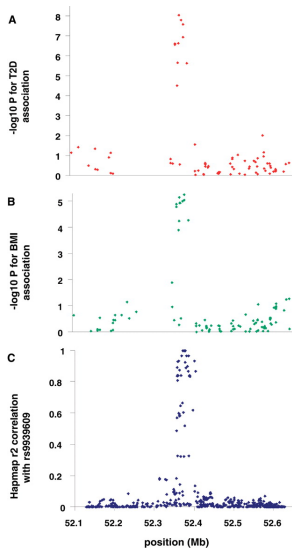
# MR in action: *FTO* → fat mass → obesity, diabetes



Using genotype as an instrumental variable, we test causality between other exposure variables and downstream phenotypes.

● Genotype in *FTO* locus → T2D

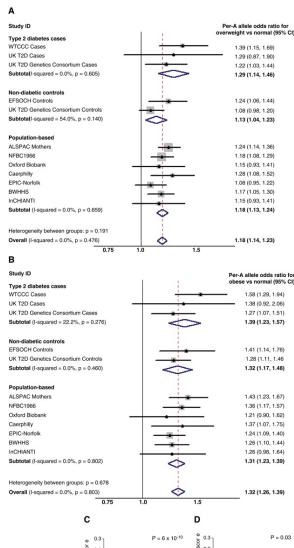
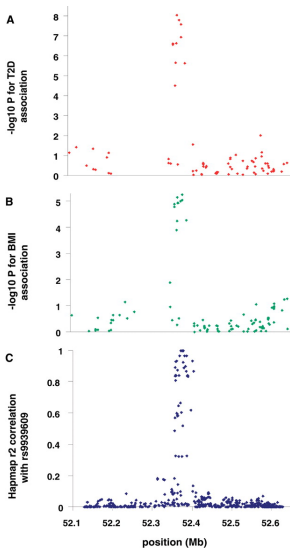
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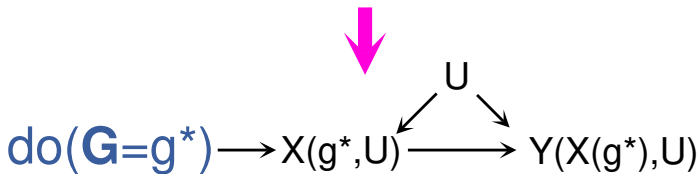
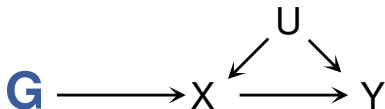
# MR in action: *FTO* → fat mass → obesity, diabetes



Using genotype as an instrumental variable, we test causality between other exposure variables and downstream phenotypes.

- Genotype in *FTO* locus → T2D
- *FTO* locus → fat mass
- Using *FTO* as "instrumental variable", we can ask other MR questions

# Why doing Mendelian Randomization?



- 1 We do not have enough knowledge of  $U$
- 2 We do not have a way to make interventions on  $\text{do}(X = x)$

*"Genotypes are beautifully randomized" - Fisher (1951)*

# Toy example: How can we learn causality between genetic disorders?

---

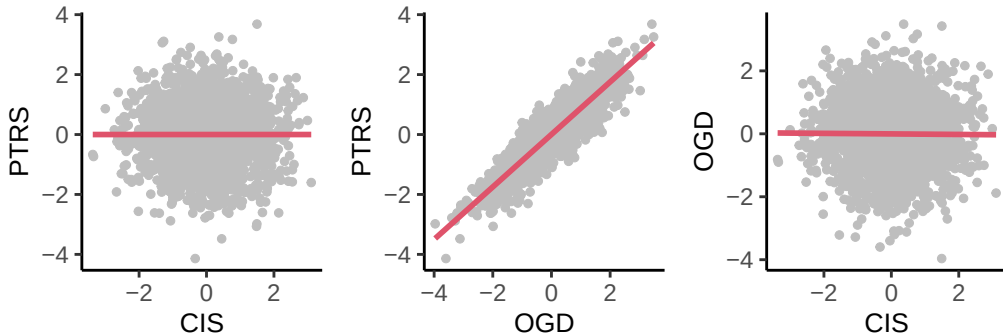
Three disorders (just for demonstration)

- OGD: Obsessive `ggplot` disorder
- PTRS: Post-traumatic R session stress syndrome
- CIS: Chronic indentation syndrome (as a result of `Python` coding)

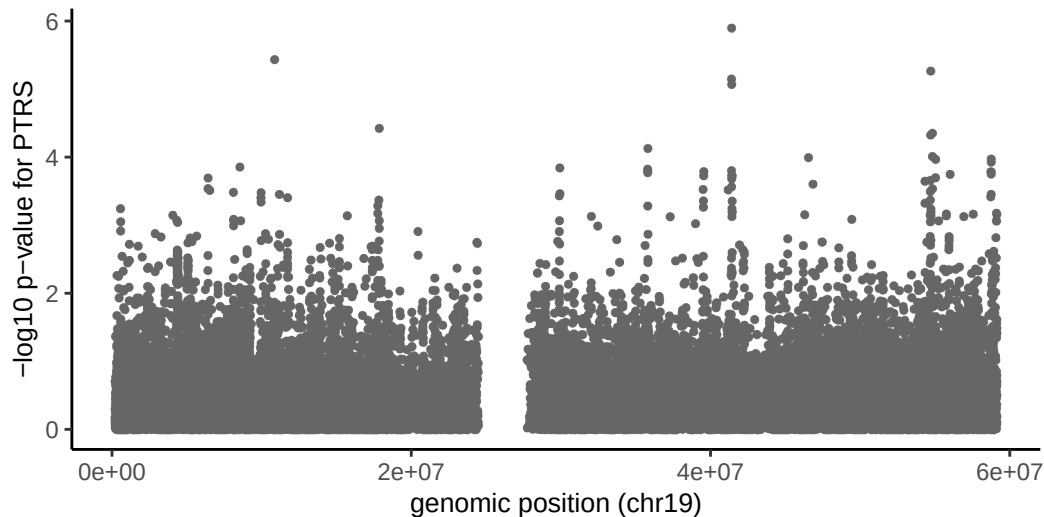
# How can we learn causality between genetic traits?

---

In previous longitudinal studies, we found some relationship.

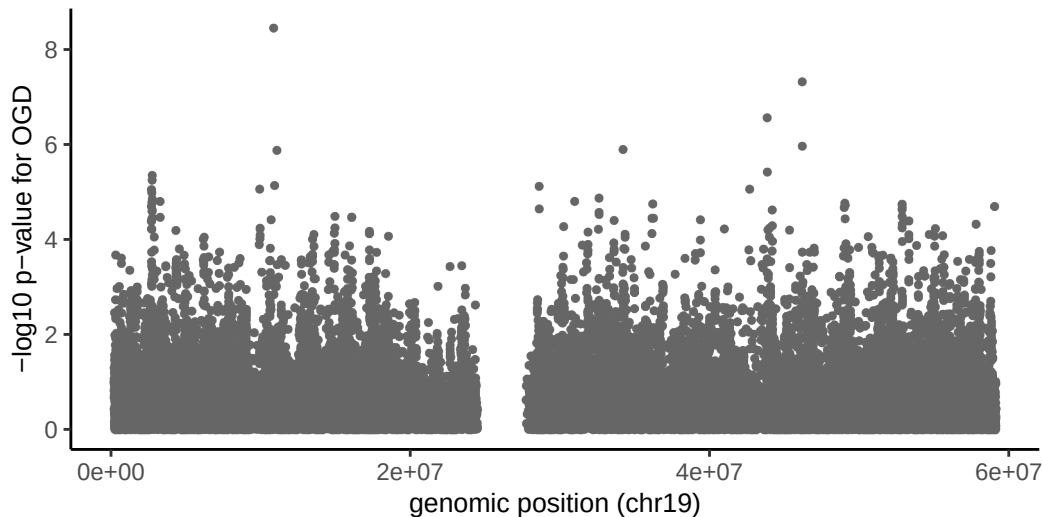


# Can we recover such relationships from GWAS data?

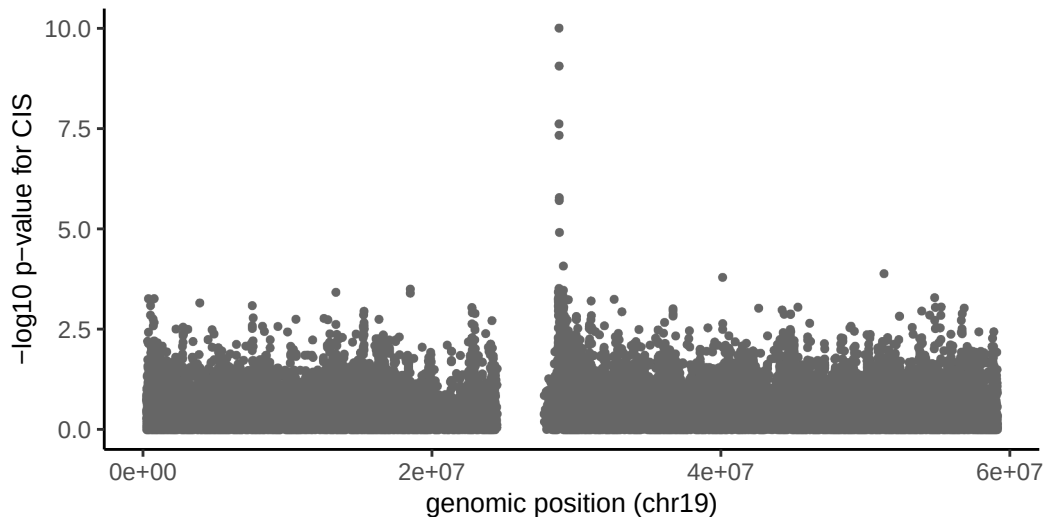




# Can we recover such relationships from GWAS data?



# Can we recover such relationships from GWAS data?



# MR measures mediation effects of $X$ to $Y$

---

Given that  $G$  is a valid instrumental variable...

*Example:*

- $X$ : gene expression
- $Y$ : disease phenotype

Suppose we have estimated

$$\begin{aligned} G &\overset{\alpha}{\rightarrow} X \\ X &= G\hat{\alpha} + \epsilon_X \end{aligned}$$

# MR measures mediation effects of $X$ to $Y$

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and

$$G \xrightarrow{\gamma} Y$$

# MR measures mediation effects of $X$ to $Y$

Given that  $G$  is a valid instrumental variable...

*Example:*

- $X$ : gene expression
- $Y$ : disease phenotype

Goal: What is the causal effect  $\beta$  in  
 $X \xrightarrow{\beta} Y$ ?

Suppose we have estimated

$$Y = X\beta + \epsilon'$$

$$\begin{aligned} G &\xrightarrow{\alpha} X \\ X &= G\hat{\alpha} + \epsilon_X \end{aligned}$$

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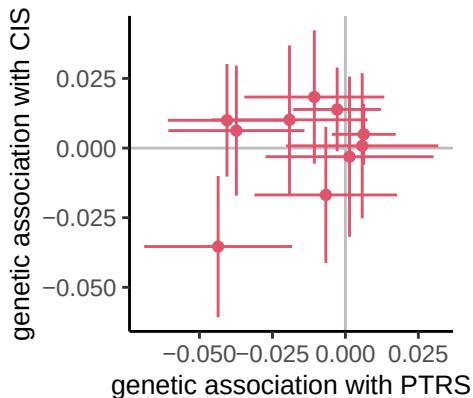
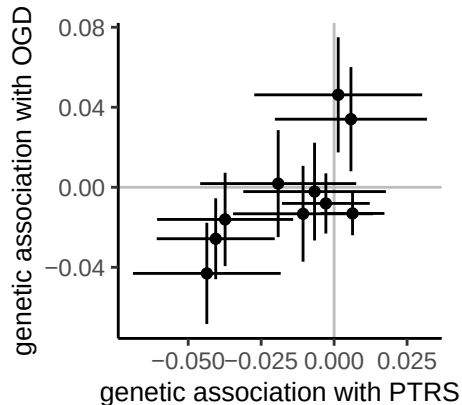
The answer is as simple as

$$\mathbb{E}[\beta] = \frac{\gamma}{\alpha}$$



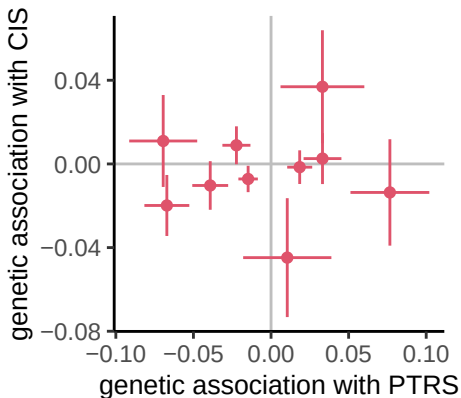
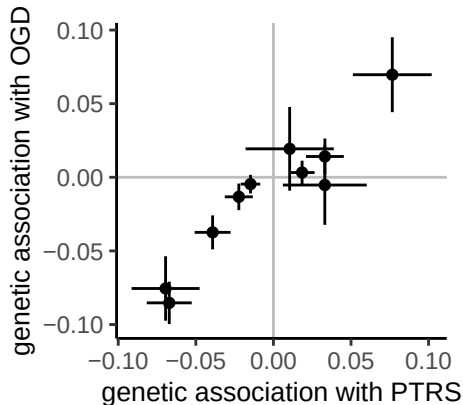
# Will any genetic variable work for MR analysis?

Goal: Is PTRS  $\rightarrow$  OGD or PTRS  $\rightarrow$  CIS?

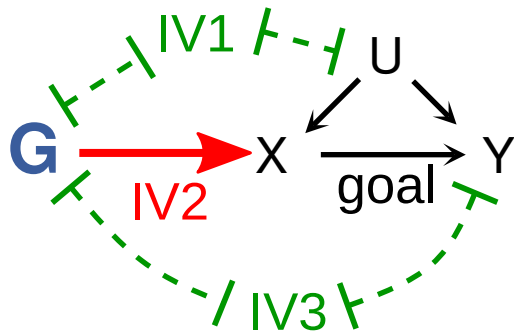


# Select causal variants of exposure (PTRS)?

Goal: Is  $\text{PTRS} \rightarrow \text{OGD}$  or  $\text{PTRS} \rightarrow \text{CIS}$ ?



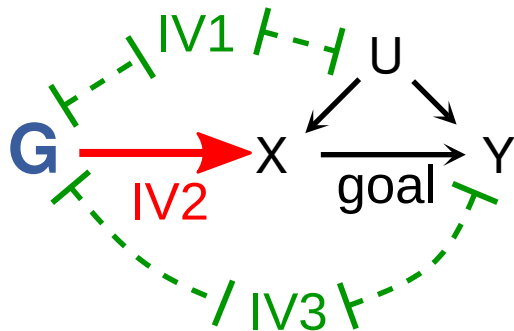
# Definition of instrumental variable in MR study



- IV1: The genetic variant  $G$  is independent of the potential confounder variable  $U$

Bowden *et al.* (2015)

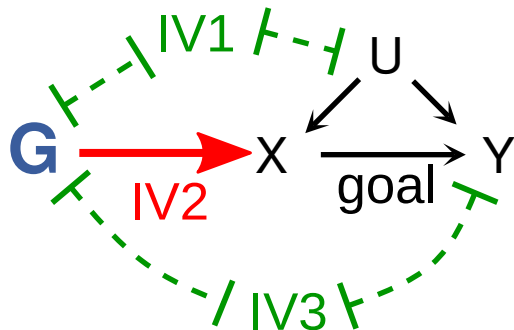
# Definition of instrumental variable in MR study



- IV1: The genetic variant  $G$  is independent of the potential confounder variable  $U$
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Bowden *et al.* (2015)

# Definition of instrumental variable in MR study

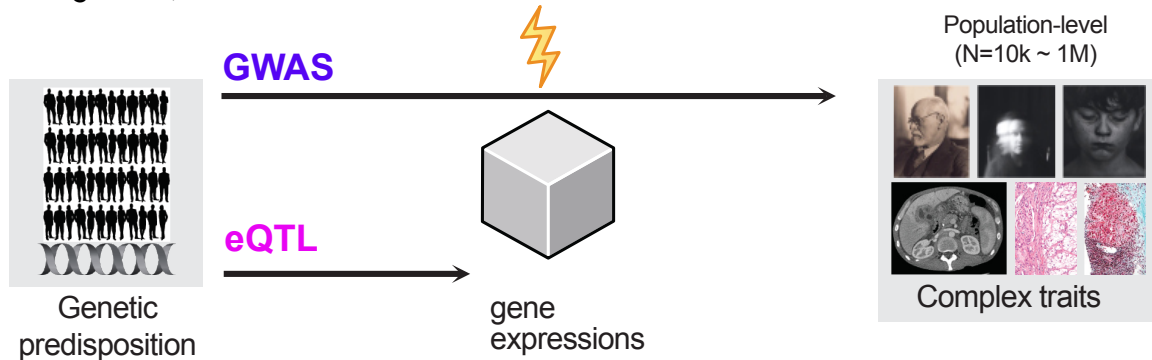


- IV1: The genetic variant  $G$  is independent of the potential confounder variable  $U$
- IV2: The genetic variant is associated with the exposure  $X$
- IV3: The genetic variant is independent of the outcome  $Y$  conditioning on  $X$

Bowden *et al.* (2015)

# Transcriptome-wide association study (TWAS)

Using eQTL, we can understand GWAS mechanisms...



$$\mathbf{m}_g \sim X\alpha_g + \epsilon_g \quad \Rightarrow \quad \mathbf{y}_{\text{GWAS}} = \sum_{g \in \text{causal genes}} \mathbf{m}_g \beta_g + \epsilon_y$$